IPL Player Performance Prediction using Machine Learning

Detailed Project Report

Abstract

The advancement of data analytics has significantly influenced modern sports, particularly cricket. The Indian Premier League (IPL) produces a large volume of structured performance data that can be analyzed to extract meaningful insights. This project presents a comprehensive machine learning-based system designed to predict IPL player performance using historical match data. The system processes ball-by-ball datasets, engineers performance features, and trains predictive models to estimate the number of runs a batsman may score in upcoming matches. Additionally, an interactive Streamlit dashboard is developed to visualize analytics such as recent form, player comparison, and explainable AI using SHAP. The project highlights how machine learning can support performance analysis, fantasy sports decision-making, and sports analytics research.

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1. Introduction

Sports analytics has become an essential component in modern cricket, enabling teams and analysts to make data-driven decisions. With the availability of detailed IPL datasets, it is possible to evaluate player performance patterns over time and predict future outcomes.
This project focuses on developing an intelligent analytics system that leverages machine learning to forecast player performance. By analyzing historical statistics such as runs scored, strike rate, venue performance, and recent form, the system generates predictions and provides explanations for those predictions. The integration of visualization and explainable AI ensures that the results are interpretable and useful for real-world applications.

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2. Background of Sports Analytics

Sports analytics involves the use of statistical methods, data mining, and machine learning to analyze sports performance. In cricket, analytics is widely used for:

- Player selection
- Match strategy planning
- Opponent analysis
- Performance tracking
- Fantasy sports recommendations

The IPL is particularly suitable for analytics due to its structured data and consistent match format.

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3. Problem Statement

Traditional evaluation of player performance relies on manual observation and simple averages, which may not capture complex performance patterns. There is a need for an automated system that can analyze large IPL datasets, identify trends, and provide reliable performance predictions along with insights explaining those predictions.

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4. Objectives

- To collect and preprocess IPL ball-by-ball and match data
- To engineer meaningful features reflecting player ability and recent form
- To build machine learning models capable of predicting player runs
- To implement explainable AI techniques for transparency
- To develop an interactive dashboard for visualization and analysis

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5. Scope of the Project

The project focuses on batsman performance prediction within IPL matches. It covers data preprocessing, feature engineering, model training, and dashboard visualization. The system is designed for educational and analytical purposes and can be extended for real-time analytics and fantasy cricket applications.

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6. Dataset Description

The dataset used includes IPL ball-by-ball deliveries and match-level information containing:

- Match ID, teams, venue, and date
- Player statistics such as runs, balls faced, and dismissals
- Contextual features including opponent and venue performance

The dataset is structured tabular data suitable for supervised machine learning.

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7. Methodology

7.1 Data Collection

Datasets were obtained from publicly available cricket analytics sources. These datasets contain multiple seasons of IPL matches, providing sufficient data for model training.

7.2 Data Preprocessing

Data preprocessing steps include:

- Cleaning inconsistent column names
- Handling missing values
- Merging deliveries and matches datasets
- Creating match-level statistics for each player

7.3 Feature Engineering

Feature engineering plays a crucial role in improving model performance. Key features created:

- Career average runs
- Previous 10 match average (recent form)
- Strike rate
- Number of boundaries (fours and sixes)
- Venue-specific performance
- Opponent-specific performance

7.4 Model Development

The project uses **XGBoost Regressor**, a gradient boosting algorithm known for strong performance on structured datasets. Hyperparameter tuning is performed to improve accuracy.

7.5 Model Evaluation

The model is evaluated using metrics such as:

- Mean Absolute Error (MAE)
- Root Mean Square Error (RMSE)

7.6 Explainable AI using SHAP

SHAP values are used to determine how each feature contributes to the final prediction. This helps users understand the reasoning behind predictions.

7.7 Dashboard Development

A Streamlit dashboard provides:

- Player selection interface
- Prediction results
- Player comparison
- Recent performance visualization
- SHAP feature importance charts

8. System Architecture

The system follows a pipeline architecture:

1. Data ingestion
2. Data preprocessing
3. Feature engineering
4. Model training
5. Prediction generation
6. Visualization through dashboard

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9. Implementation Details

The project is implemented using Python and several data science libraries. Scripts are organized for preprocessing, feature engineering, model training, and dashboard deployment. The trained model is saved and reused by the Streamlit application for real-time predictions.

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10. Results and Discussion

The developed model successfully predicts approximate runs for selected players using historical performance patterns. The dashboard enables interactive analysis and demonstrates how recent form significantly influences predictions. SHAP visualizations reveal that features such as strike rate and previous match averages are strong predictors of performance.

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11. Applications

- Fantasy cricket team selection
- Player scouting and performance tracking
- Match strategy planning
- Sports analytics research
- Educational machine learning projects

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12. Advantages

- Automated performance prediction
- Interactive visualization
- Explainable AI integration
- Scalable architecture
- Real-world applicability

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13. Limitations

- Predictions rely on historical data
- External factors such as injuries and weather are not included
- New players with limited data may produce less accurate predictions

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14. Future Scope

- Bowler performance prediction
- Match winner prediction
- Dream11 team recommendation system
- Real-time data integration via APIs
- Deep learning models (LSTM)
- Mobile application development
- Deployment as a public web analytics platform

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15. Conclusion

This project demonstrates the effective use of machine learning in sports analytics for predicting IPL player performance. By combining data preprocessing, feature engineering, predictive modeling, and explainable AI, the system provides meaningful insights alongside predictions. The Streamlit dashboard enhances usability and allows interactive exploration of player analytics. The project highlights the importance of data-driven decision-making in modern cricket and provides a foundation for advanced sports analytics solutions.

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16. References

- Kaggle IPL datasets
- XGBoost official documentation
- SHAP documentation
- Streamlit documentation
- Research articles on sports analytics

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17. Appendix

Project files include preprocessing scripts, feature engineering modules, trained models, and the Streamlit dashboard application.